

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Completed structural analysis for reinforced concrete and steel elements using **GSA**, reducing design iteration time by **15%** through efficient model setup and simulation.
- Prepared detailed mark-ups and collaborated with structural technicians to produce intricate structural drawings in **Revit**, ensuring adherence to project specifications and constructability requirements.
- Represented Arup in client and consultant meetings, effectively communicating complex structural design concepts and contributing to project strategy discussions, leading to a **90% stakeholder satisfaction** rating in project reviews.
- Conducted thorough site visits and structural investigations for existing buildings, identifying critical structural integrity issues and providing actionable remediation plans to prevent potential failures, mitigating risk by **30%**.
- Authored comprehensive technical reports and site investigation summaries, clearly articulating findings and recommendations to both technical and non-technical audiences, improving project documentation clarity by **20%**.
- Collaborated with cross-functional Design Teams on managing project data within Common Data Environments like **ACC**, facilitating seamless information exchange and enhancing project delivery efficiency.

SKILLS

Structural Analysis & Design – GSA, Tekla Structural Designer, Reinforced Concrete, Steel, Timber, Masonry, Structural Investigations

BIM & Drafting – Revit, ACC (Common Data Environment)

Project Management & Communication – Client Meetings, Consultant Collaboration, Report Writing, Presentations, Site Visits

Software & Tools – Python, Java, SQL, Spring Boot, AWS, Linux, Jenkins, Jira, Confluence, Power BI, Tableau

Data Analytics & BI – Data Modelling, DAX, Power Query, Statistical Analysis, R, pandas, NumPy

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Structural Integrity Assessment & Retrofit Design Platform

- Developed a Python-based analysis tool to process structural investigation data from site visits, integrating findings from **GSA** and **Revit** models to predict potential failure points with **85% accuracy**.

- Designed and documented retrofit solutions for identified structural weaknesses in concrete and steel elements, preparing detailed specifications for construction teams and estimating material requirements, reducing projected repair costs by **22%**.
- Authored comprehensive reports detailing structural assessments and proposed retrofit designs, presented findings to clients and project stakeholders, securing approval for implementation in **5 major infrastructure projects**.

Automated Financial Transaction Monitoring Dashboard

- Engineered a Python-based ETL process to ingest and transform high-volume transaction data from PostgreSQL and Oracle sources, populating a financial-services dashboard for operational alerting.
- Developed real-time alerts for failed or stalled payment messages, reducing manual intervention by **40%** and improving incident response times within the payments operations dashboard.
- Implemented data validation and reconciliation workflows using SQL queries, ensuring data integrity and supporting audit trail requirements for a critical financial-services client.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)